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ARTICLE

Thermal Performance of Intermediate Heat Exchanger during High-Temperature Continuous Operation in HTTR

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To use the High-Temperature Gas-cooled Reactor (HTGR) system as a commercial reactor similar to the future HTGR 'GTHTR300C,' supplying stable high-temperature heat from the reactor to the heat utilization system such as the hydrogen production system over a prolonged period should be demonstrated. The 50-day high-temperature continuous operation with full power, which is the first long-term operation with a reactor outlet coolant temperature over 900°C, was achieved in March 2010 in the High Temperature Engineering Test Reactor (HTTR) of the Japan Atomic Energy Agency (JAEA). In order to supply stable high-temperature heat from the reactor to the heat utilization system using the HTTR, the heat exchange performance and structure integrity of the intermediate heat exchanger (IHX) should be confirmed. In this paper, the heat exchange performance and structure temperature of the IHX were evaluated with the high-temperature continuous operation data and compared with the designed one or estimated one. As a result, it was confirmed that the IHX can exchange stable high-temperature heat from the primary coolant to the secondary helium, and the IHX structure temperature is under the allowable working temperature in the operation. Moreover, it was confirmed that the heat exchange performance is almost the same as the designed one and that the structure temperature estimated with the analysis code is almost the same as the measured one.

KEYWORDS: HTGR, HTTR, high-temperature continuous operation, intermediate heat exchanger, heat exchange performance, structural integrity

I. Introduction

The High Temperature Engineering Test Reactor (HTTR),[1] which is the first High-Temperature Gas-cooled Reactor (HTGR) in Japan, having a thermal power of 30 MW and 950°C maximum reactor outlet coolant temperature, was constructed in the Japan Atomic Energy Agency (JAEA) for the purpose of establishing and improving the technologies of HTGRs as well as nuclear heat utilization. The HTTR attained its first criticality in November 1998, and in-service operation has been conducted through five-phase rise-to-power tests.

To develop the commercial HTGR system such as the future HTGR 'GTHTR300C,' supplying stable heat to a heat utilization system over a prolonged period should be demonstrated. Thus, the long-term continuous operation with full power and over 900°C reactor outlet coolant temperature (called 'high-temperature continuous operation') was planned.

Before the high-temperature continuous operation, the long-term continuous operation with 850°C maximum reactor outlet coolant temperature mode was conducted from March to April in 2007[2] in order to verify and confirm the safety of the high-temperature continuous operation.

The high-temperature continuous operation was started from January 2010, and the 50-day high-temperature continuous operation with full power was achieved in March 2010.[3] As a result of these operations, it was demonstrated that the HTTR can supply stable heat to the heat utilization system such as the thermochemical hydrogen production system over a prolonged period. These operations are the first achievements in the history of the HTTR. The data obtained, such as core burnup characteristics, helium purity management, high-temperature instrument performance, and core internal structure integrity, are very useful for the development of a commercial HTGR.

The intermediate heat exchanger (IHX) is one of the high-temperature instruments installed in the HTTR. It is planned that the thermochemical hydrogen production system is connected to the HTTR through the IHX, so the IHX is the most important instrument in the development of a hydrogen production system and the future HTGR 'GTHTR300C.'

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Table 1 Major specifications of the HTTR

Thermal power	30 MW
Fuel	UO_2
Uranium enrichment	3–10% (average 6%)
Fuel assembly type	Pin-in-block type
Core	
Structure material	Graphite
Height	2.9 m
Diameter	2.3 m
Number of main cooling loop	1
Coolant	Helium
Primary coolant pressure	4 MPa
Reactor inlet coolant temperature	395°C
Reactor outlet coolant temperature	950°C

In this paper, the heat exchange performance and structure temperature of the IHX for the high-temperature continuous operation were evaluated and compared with the designed data or estimation result.

II. Outline of the HTTR and the High-Temperature Continuous Operation

1. Outline of the HTTR

(1) Reactor and Fuel

Table 1 shows the major specifications of the HTTR and **Fig. 1** shows the structure of the reactor of the HTTR. The reactor is composed of fuel elements, reflector block, core restraint mechanism, reactor pressure vessel (RPV), and so on. The reactor core is composed of 30 columns of the fuel blocks and 7 columns of the control rod guide blocks. The permanent reflector blocks and the replaceable reflector blocks are arranged around the reactor core.

Figure 2 shows the components of the fuel element. The fuel element of the HTTR is the so-called pin-in-block type composed of fuel rods and a hexagonal graphite block (fuel block). The fuel block has 31 or 33 holes, and the fuel rods are inserted in the hole. The primary coolant flows through the annulus formed between the fuel rod and hole of the fuel block.

In the reactor, the primary coolant flows in from the annulus of the primary concentric hot gas duct arranged at the bottom of the RPV. It rises to the upper plenum through the gap between the RPV and the permanent reflector blocks. Next, it turns over and flows downward through the fuel channel. Lastly, heated coolant is joined at the hot plenum and flows out to the inner pipe of the primary concentric hot gas duct through the primary coolant outlet nozzle.

(2) HTTR Cooling System

Figure 3 shows the HTTR cooling system. The HTTR cooling system is composed of the main cooling system (MCS), the auxiliary cooling system (ACS), and the vessel cooling system (VCS). The MCS is operated in normal operation to remove the generated heat at the reactor.

The ACS and VCS are the engineered safety features. The ACS is in standby during normal operation and starts up to remove residual heat after a reactor scram. The VCS cools the biological concrete shield surrounding the RPV at nor-

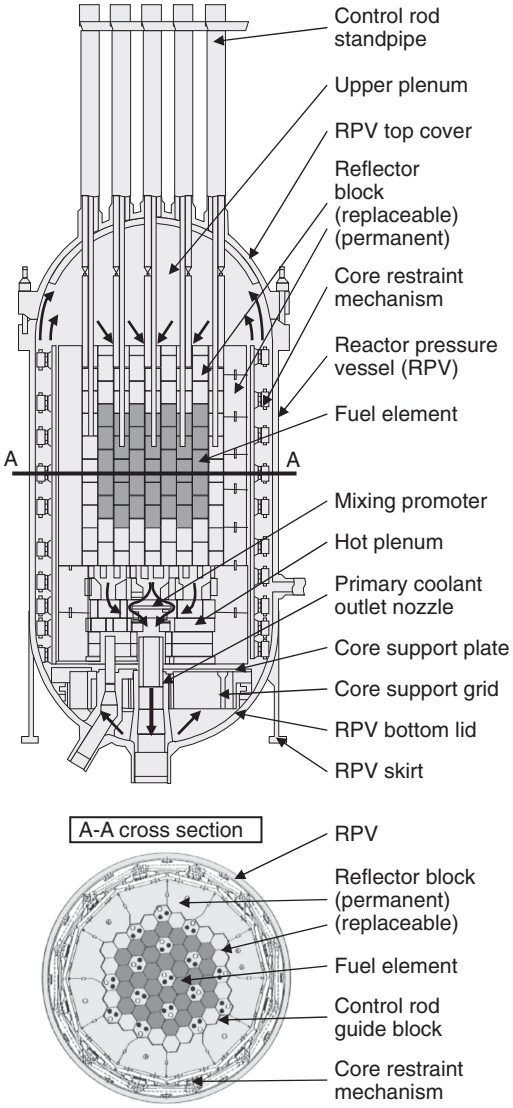


Fig. 1 Structure of the HTTR reactor

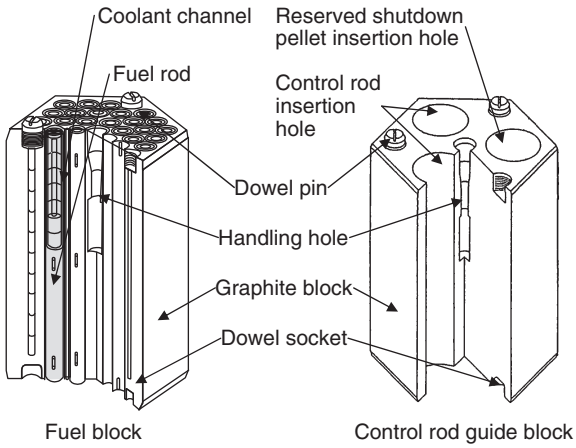


Fig. 2 Component of fuel element

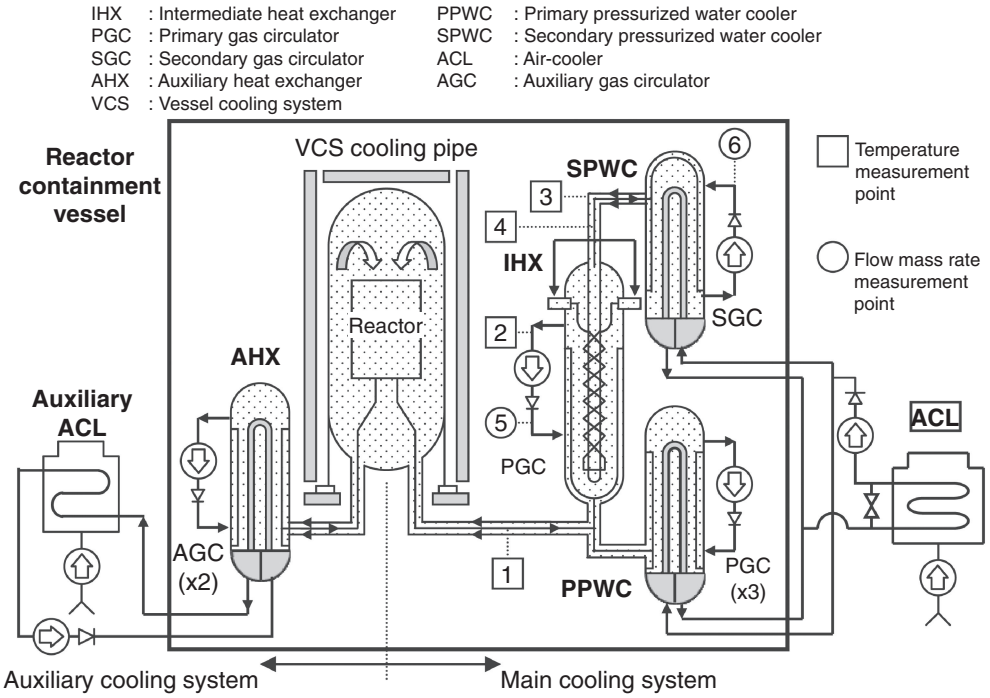


Fig. 3 HTTR main cooling system

mal operation, and removes heat from the core by natural convection and thermal radiation outside the reactor pressure vessel under 'accident without forced cooling' conditions such as a rupture of the primary concentric hot gas duct, when neither the MCS nor the ACS can cool the core effectively.

The MCS is composed of the primary cooling system, the secondary cooling system, and the pressurized water cooling system. The heat generated at the reactor core is dispersed into the atmosphere by way of these cooling systems. The HTTR has the following primary missions: (1) establish heat utilization technology using heat generated at the reactor, and (2) demonstrate the inherent safety features of HTGR. To achieve mission (1), the 10 MW high-temperature gas-gas-type heat exchanger is needed. Also, to achieve mission (2), the 30 MW heat exchanger is needed. Thus, the heat exchanger of one IHX and one primary pressurized water cooler (PPWC) are installed in the primary cooling system and connected to the reactor with the primary coolant concentric hot gas duct in the HTTR. The heat exchanger of one secondary pressurized water cooler (SPWC) is installed in the secondary cooling system and connected to the IHX with the secondary helium concentric hot gas duct. The heat exchanger of one air-cooler is installed in the pressurized water cooling system. The primary gas circulators (PGCs) are arranged around the IHX and PPWC; the secondary gas circulator (SGC) is arranged around the SPWC. The heated primary coolant at the reactor flows inside the inner tube of the primary concentric hot gas duct to the IHX and PPWC. The primary coolant is cooled to about 400°C at the IHX and PPWC, and returns to the reactor through the annular space between the inner pipe and outer pipe of the primary concentric hot gas duct.

The HTTR has two reactor outlet coolant temperature modes and two systematical operation modes. One of the reactor outlet coolant temperature modes is called the "Rated operation mode" with a maximum reactor outlet coolant temperature of 850°C; the other is the "High-temperature test operation mode" with a maximum reactor outlet coolant temperature of 950°C. In these operation modes, the reactor outlet coolant temperature is changed by changing the primary coolant mass flow rate.

The HTTR has two systematical operation modes of "Single-loaded operation mode" and "Parallel-loaded operation mode." The generated heat of 30 MW at the reactor is removed using only the PPWC in the single-loaded operation mode. On the other hand, the generated heat is removed using both the PPWC and IHX in the parallel-loaded operation mode; the amounts of heat removed are 20 MW at the PPWC and 10 MW at the IHX. Thus, the HTTR operation mode is generally called the "Rated/Single-loaded operation."

(3) IHX

Table 2 shows the major specifications of the IHX and **Fig. 4** shows the structure of the IHX. The IHX is the vertical helically-coiled counterflow-type gas-gas heat exchanger. The body of the IHX is composed of the inner shell and outer shell. A thermal insulator is arranged inside of the inner shell. **Figure 5** shows the structure around the heat transfer tube in the IHX. The IHX has 96 heat transfer tubes arranged with 6 lamination layer structures. The heat transfer enhancement plates are installed between the heat transfer tubes in order to enhance the heat exchange performance (*i.e.*, the heat is exchanged with convective heat transfer and thermal radiation due to the heat transfer enhancement plates in the IHX).

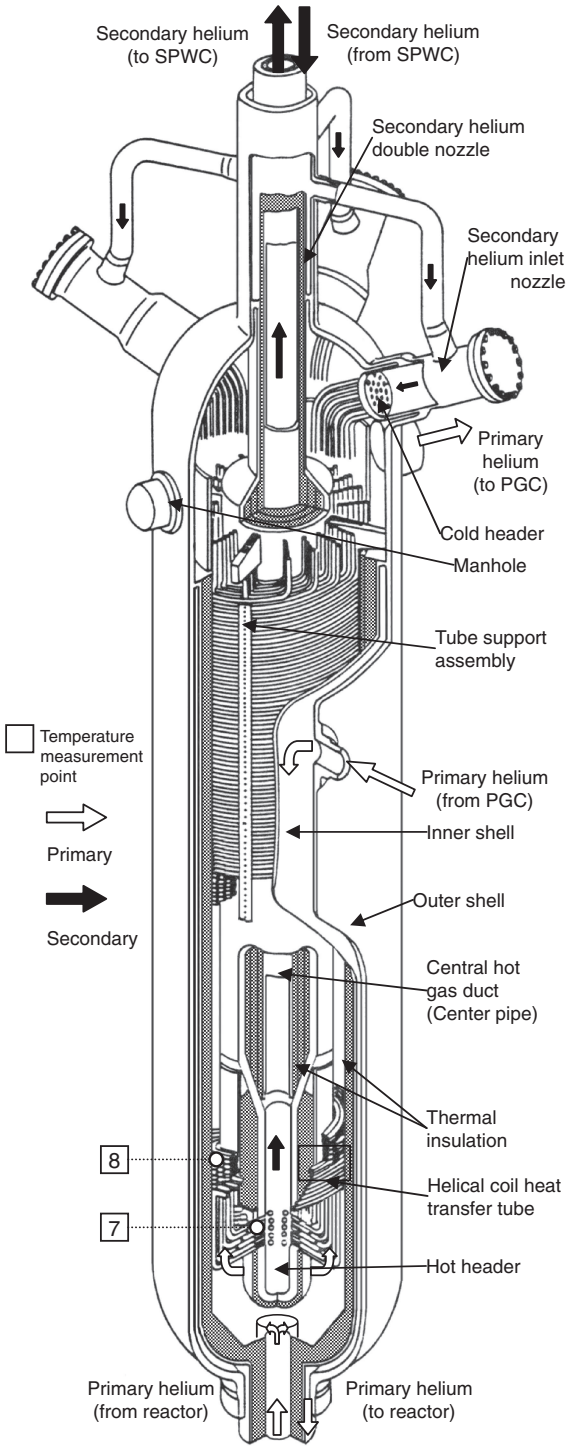


Fig. 4 Structural diagram of the IHX

The hot primary coolant enters into the IHX through the inner tube of the primary concentric hot gas duct. It is deflected under the hot header and discharged around the heat transfer tubes to transfer the heat to the secondary helium cooling system. It flows to the PGC via the upper outlet nozzle and flows back to the annular space between the inner shell and outer shell.

On the other hand, the secondary helium gas flows downwards in the heat transfer tubes and upwards in the central hot gas duct through the hot header. The inner thermal insulator is installed inside the inner shell to maintain its temperature under the allowable one. The thermal insulators are installed outside and inside of the central hot gas duct. The heat transfer is reduced by them at the central hot gas duct so that high efficiency can be obtained. Moreover, the central hot gas duct temperature is kept under the allowable working temperature by them. The pressure of the secondary helium is controlled higher than that in the primary coolant because of the prevention of the primary helium leak due to the rupture of the heat transfer tube in the IHX.

The tube support assemblies hold the heat transfer tube. Both the central hot gas duct and heat transfer tube support assemblies are hung from the vessel top so that the thermal expansion is not constrained.

Table 2 Major specifications of the IHX

Heat exchanger capacity	10 MW
IHX primary/secondary helium gas	
Flow rate	12 t/h/12 t/h
Inlet temperature	950°C/300°C
Outlet temperature	390°C/860°C
Heat transfer tube	
Number	96
Outer diameter	31.8 mm
Thickness	3.5 mm
Outer shell outer diameter	2.0 m
Total height	11.0 m
Material	
Heat transfer tube	Hastelloy XR
Hot header	Hastelloy XR
Hot duct	Hastelloy XR
Outer/Inner shell	Low-alloy steel
Allowable working pressure	
Outer shell	4.8 MPa
Heat transfer tube	0.3 MPa[a)]
Allowable working temperature	
Outer shell	430°C
Heat transfer tube	955°C

[a)] differential pressure

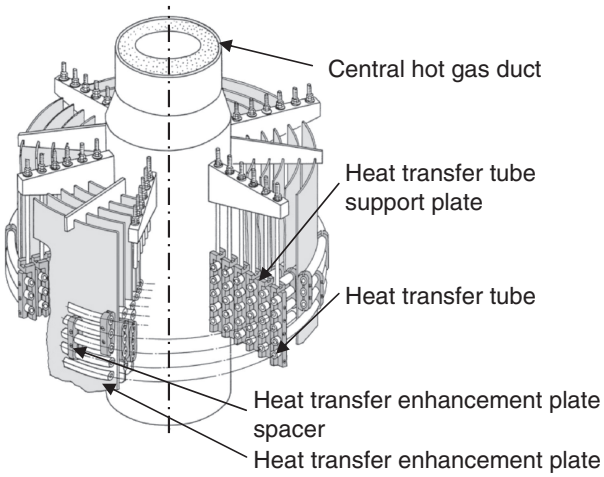


Fig. 5 Structure around the heat transfer tube in the IHX

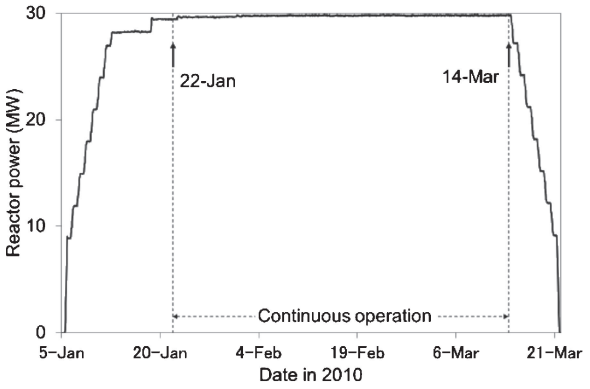


Fig. 6 Historical trend during the high-temperature continuous operation

2. Outline of the High-Temperature Continuous Operation

The high-temperature continuous operation (called HP-11) started from 17 December 2009 to 21 March 2010.[3] The reactor was started up on 17 December 2009 and shut down on 27 December 2009 for the purpose of measurement of control rod worth.

Figure 6 shows the historical trend during the high-temperature continuous operation. The reactor was started up on 5 January 2010. The reactor power was raised gradually and reached to about 98% (29.5 MW) on 22 January 2010. After the thermal power calibration, the reactor power was raised to about 99% (29.7 MW) on 31 January 2010. The 50-day high-temperature continuous operation with full power was achieved from 22 January 2010 to 13 March 2010. The reactor power decrease was started on 14 March 2010 and the reactor was shut down on 21 March 2010. The 69 effective full-power operation days were added in the operation.

III. Evaluation of Heat Exchange Performance and Structure Temperature of the IHX

1. Evaluation Method

(1) Heat Exchange Performance of the IHX

The heat exchange performance of the IHX was evaluated with the high-temperature 50-day continuous operation data. The results were compared with previous operation data to confirm the degradation of the performance.

Generally, exchanged heat at the heat exchanger is given by the following equation.

$$Q = UA\Delta T \tag{1}$$

The UA in this equation, which is the product of heat transmission coefficient U and heat transfer area of the heat exchanger A, is the characteristic value of each heat exchanger and it shows the heat exchange performance. Thus, the heat exchange performance of the IHX was evaluated with the UA. The UA of the IHX is given by the following equation, which is given by the transformation of Eq. (1).

$$UA = Q_{\mathrm{IHX}}/\Delta T_{\mathrm{IHX}} \tag{2}$$

Table 3 Measurement equipment and measurement accuracy

No.	Measurement point (Measurement equipment)	Range	Accuracy
1	Reactor outlet primary coolant temperature (K-type thermocouple)	0–1,200°C	±0.4%
2	IHX outlet primary coolant temperature (K-type thermocouple)	0–500°C	±0.4%
3	IHX inlet secondary coolant temperature (K-type thermocouple)	0–500°C	±0.4%
4	IHX outlet secondary coolant temperature (K-type thermocouple)	0–1,000°C	±0.4%
5	IHX primary coolant flow mass rate (D/P-type flow meter)	0.0–17.0 t/h	±0.5%
6	SPWC secondary coolant flow mass rate (D/P-type flow meter)	0.0–17.0 t/h	±0.5%
7	IHX hot header ligament temperature (K-type thermocouple)	0–1,000°C	±0.4%
8	IHX heat transfer tube temperature (K-type thermocouple)	0–1,000°C	±0.4%

The logarithmic mean temperature difference for the IHX is given by the following equation.

$$\Delta T_{\mathrm{IHX}} = \frac{T^{\mathrm{P}}_{\mathrm{in}} - T^{\mathrm{S}}_{\mathrm{out}} - (T^{\mathrm{P}}_{\mathrm{out}} - T^{\mathrm{S}}_{\mathrm{in}})}{\ln\left[\dfrac{T^{\mathrm{P}}_{\mathrm{in}} - T^{\mathrm{S}}_{\mathrm{out}}}{T^{\mathrm{P}}_{\mathrm{out}} - T^{\mathrm{S}}_{\mathrm{in}}}\right]} \tag{3}$$

The heat exchanged at the IHX is considered as the heat received at the IHX given by the following equation.

$$Q_{\mathrm{IHX}} = Q^{\mathrm{S}}_{\mathrm{IHX}} = C^{\mathrm{S}}_{\mathrm{P}} G^{\mathrm{S}}_{\mathrm{IHX}} (T^{\mathrm{S}}_{\mathrm{out}} - T^{\mathrm{S}}_{\mathrm{in}}) \tag{4}$$

Table 3 shows the measurement point used in the present evaluation. The numbers of measurement points shown in Table 3 correspond to that shown in Figs. 3 and 4. The plant is most stable at 7:00 am–7:59 am in a day, so the average data obtained during that interval was used in the evaluation.

(2) IHX Structure Temperature

The IHX structure temperature obtained in the high-temperature continuous operation is compared with result estimated using the estimation code. The structure and coolant temperatures in the IHX were estimated using the IHX steady-state estimation code, which is used to solve the heat balance equation. The code is constructed with a two-dimensional cylindrical coordinate thermal network model shown in **Fig. 7**. Thermal conductivity, convective heat transfer between the structures and the coolant, thermal radiation between the structures, and heat transport by coolant flow were considered in the code.

The Nusselt number outside of the heat transfer tube (*i.e.*, primary coolant side) is based on the following equation.[4,5]

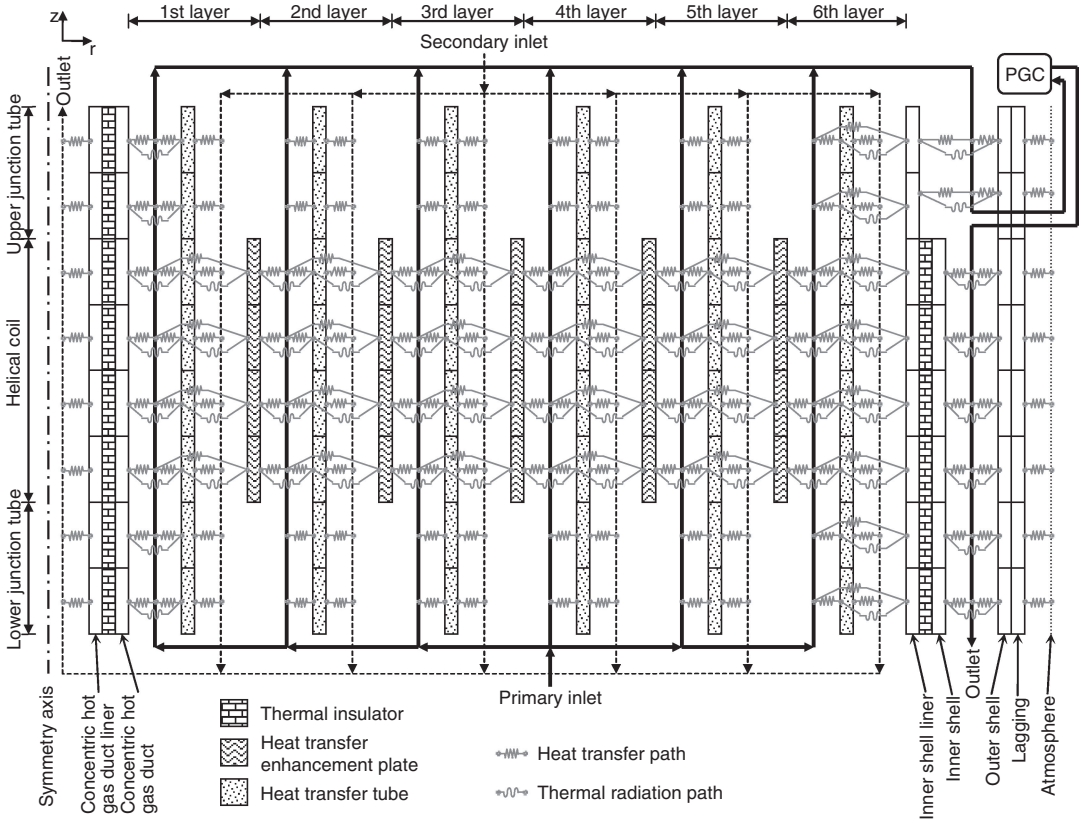


Fig. 7 Estimation model for IHX temperature estimation

$$Nu_{out} = 0.057 \cdot C_0 \cdot Re^{0.8} \cdot Pr^{0.4} \quad (800 < Re \leq 7{,}000)$$
$$= 0.33 \cdot C_0 \cdot Re^{0.6} \cdot Pr^{0.3} \quad (7{,}000 < Re) \tag{5}$$

$$C_0 = \eta_f \cdot \eta_e \cdot \eta_\theta \cdot \eta_s \cdot E \tag{6}$$

The Nusselt number on the heat enhancement plate surface is based on the following Nusselt number equation.[6)]

$$Nu_{enh} = C_1 \cdot Re^{0.54} \cdot Pr^{0.4} \cdot \left(\frac{X_w}{d_{out}} - 1\right)^{-0.47} \cdot \left(\frac{P_a}{d_{out}} - 1\right)^{0.5} \tag{7}$$

The Nusselt number inside of the heat transfer tube (*i.e.*, secondary coolant side) is based on the following equation.[4,5,7)]

$$Nu_{in} = \left(\frac{0.148\sqrt{D}}{1 - 1.813\sqrt{D}}\right) \cdot 0.021 \cdot Re^{0.8} \cdot Pr^{0.4} \quad \text{(Laminar region)}$$
$$= 0.0228 \cdot Re^{5/6} \cdot Pr^{0.4} \cdot \left(\frac{d}{D}\right)^{1/12} \cdot \left\{1 + \frac{0.0732}{[Re \cdot (d_{in}/D_h)^{2.5}]^{1/6}}\right\} \quad \text{(Turbulent region)} \tag{8}$$

$$D = Re \cdot \sqrt{\frac{d_{in}}{D_h}} \tag{9}$$

The convective heat transfer and thermal radiation are modeled as the thermal path shown in Fig. 7. The primary coolant/secondary helium IHX inlet temperature and flow rate were given as the boundary condition.

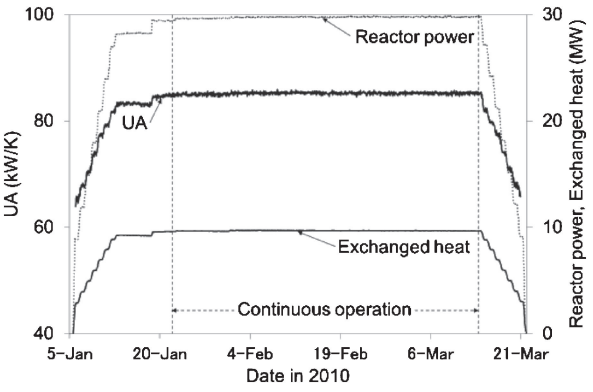


Fig. 8 Historical trend of the reactor power, exchanged heat, and *UA* of the IHX

2. Evaluation Result

(1) Heat Exchange Performance

To supply stable high-temperature heat from the reactor to the heat utilization system, it is necessary to confirm a stable heat exchange at the IHX. Thus, the stabilities of the supplying heat, supplying temperature, and heat exchange performance were confirmed.

Figure 8 shows the historical trend of the reactor power, exchanged heat at the IHX, and *UA* of the IHX in the high-

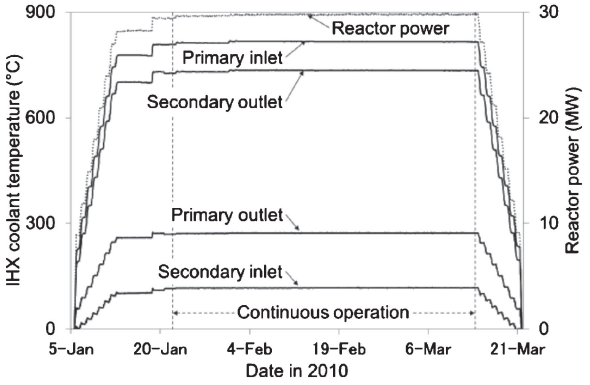


Fig. 9 Historical trend of the primary coolant and secondary helium gas temperature at the IHX

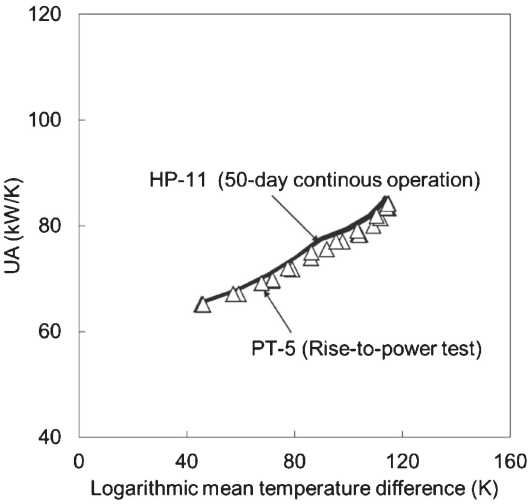


Fig. 10 Relationship between the *UA* and logarithmic mean temperature difference on the IHX

temperature continuous operation. The exchanged heat at the IHX and *UA* of the IHX were almost constant throughout the operation period. Also, the exchanged heat was almost 10 MW, which is 1/3 of the generated heat at the reactor and design value.

Figure 9 shows the historical trend of the primary coolant and secondary helium temperature at the IHX. The high-temperature primary coolant was supplied to the IHX, and then the high-temperature secondary helium was generated and supplied to the secondary helium cooling system.

Figure 10 shows the relationship between the *UA* and logarithmic mean temperature difference on the IHX. In this figure, the result for the high-temperature continuous operation and previous high-temperature test operation[8] (performed in 2004) is shown.[9] Both results are almost the same, so the degradation of the heat exchange performance is not confirmed. It is confirmed that the IHX can exchange the stable high-temperature heat from the primary coolant to the secondary helium.

(2) IHX Structure Temperature

Figure 11 shows the comparison of the measured temperature with the estimated one at the IHX heat transfer tube

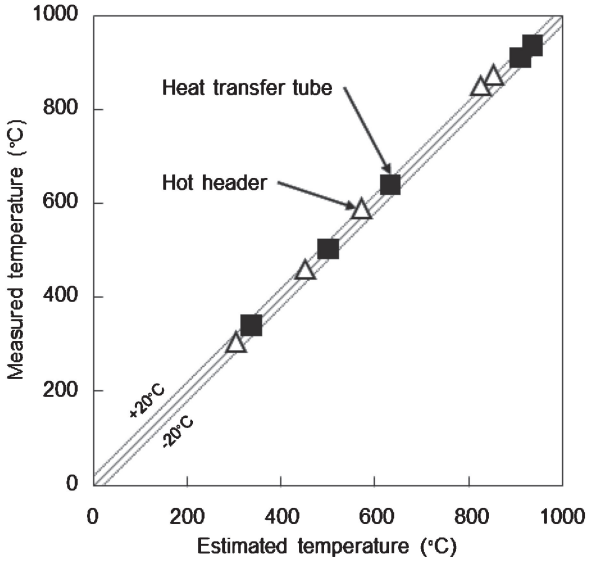


Fig. 11 Comparison of the measured temperature with estimated one at IHX heat transfer tube and hot header

and hot header. The difference between the estimated temperature and the measured one becomes larger as the temperature increases at the IHX heat transfer tube and the hot header. Under the condition of IHX inlet coolant temperature over 900°C, the estimated temperature at the heat transfer tube is about 5°C higher than the measured one; this value is about 0.5% of the measured one. The estimated temperature at the hot header is about 20°C lower than the measured one; this value is about 2% of the measured one.

The temperature difference is caused by the modeling of the hot header. The structure near the measurement point of the hot header is very complex, which is composed of the hot header and the lower junction tube. The estimation model does not represent these complex structures in detail.

Thus, the difference between the measured temperature and the estimated one can be identified, and it can be considered that this estimation method can be applied to predict the temperature for a vertical helically coiled gas-gas heat exchanger.

3. Development of Design for Future HTGR 'GTHTR300C'

In order to demonstrate the hydrogen production with the heat generated at the reactor, it is planned that the HTTR is connected to a hydrogen production system through the IHX and secondary helium cooling system (called as "HTTR-IS"). The conceptual design of the HTTR-IS system is ongoing in JAEA. It is planned that the temperature of the secondary helium gas being supplied to the hydrogen production system (*i.e.*, secondary helium IHX outlet temperature) is set to about 900°C by the reduction of the secondary helium gas flow rate.

Moreover, in order to make practical use of HTGR, a future HTGR system 'GTHTR300C' producing electric power and hydrogen is researched and developed in JAEA. It is planned that a vertical helically-coiled gas-gas heat exchanger like the IHX is adopted in the GTHTR300C.[10]

It is very important to estimate the temperature of the internal structure in the heat exchanger with high accuracy because the temperature of the internal structure is high and the heat transfer tube is a part of the primary coolant pressure boundary. The results such as the heat exchange performance and estimation of the IHX structure temperature are very useful for these research and development efforts.

IV. Conclusions

The 50-day high-temperature continuous operation with full power was achieved in March 2010 in the HTTR. In this paper, the heat exchange performance and structure temperature of the IHX were evaluated with the high-temperature continuous operation data and compared with the designed one or estimated one.

As a result, the following facts were confirmed:

(1) The IHX can exchange stable high-temperature heat from the primary coolant to the secondary helium in the operation.
(2) The structure temperature is under the allowable working temperature in the operation.
(3) The heat exchange performance is almost the same as the designed one.
(4) The structure temperature estimated using the analysis code is almost the same as the measured one; the difference is about 2%.

The results are very useful for the research and development of the HTTR-IS and the future HTGR.

Nomenclature

A: Heat transfer area of the IHX (m^2)
C_1: Coefficient based on the definition mentioned in the thesis[6] (—)
C_P^S: Secondary helium specific heat (J/kg·K)
D_h: Diameter of helical coil (m)
d_{in}: Inner diameter of heat transfer tube (m)
d_{out}: Outer diameter of heat transfer tube (m)
E: Coefficient on the heat transfer enhancement effect (—)
G_{IHX}^S: Secondary helium mass flow rate at the IHX (kg/s)
Nu_{enh}: Nusselt number on heat transfer enhancement plate (—)
Nu_{in}: Nusselt number inside of heat transfer tube (—)
Nu_{out}: Nusselt number outside of heat transfer tube (—)
P_a: Pitch of heat transfer tube in flow direction (m)
Pr: Prandtl number (—)
Q_R: HTTR reactor thermal power (W)
Q_{IHX}: Exchanged heat at the IHX (W)
Q_{IHX}^S: Secondary helium heat received at the IHX (W)
Re: Reynolds number (—)
T_{in}^P: Primary helium IHX inlet temperature (°C)
T_{out}^P: Primary helium IHX outlet temperature (°C)
T_{in}^S: Secondary helium IHX inlet temperature (°C)
T_{out}^S: Secondary helium IHX outlet temperature (°C)
U: Heat transmission coefficient of the IHX ($W/m^2 \cdot K$)
X_w: Width of flow path between heat transfer enhancement plates (m)
ΔT: Logarithmic mean temperature difference (°C)
ΔT_{IHX}: Logarithmic mean temperature difference of the IHX (°C)
η_f: Coefficient on degradation caused by deformation of heat transfer tube at bending process (—)
η_e: Coefficient on degradation caused by decentering of helically-coiled tube (—)
η_θ: Coefficient on skew flow at helically-coiled heat transfer tube (—)
η_s: Coefficient on the pitch of heat transfer tube in flow direction (—)

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